



EcoCreds: AI Platform for Shoppers

Final Year Project Report | ICS-851

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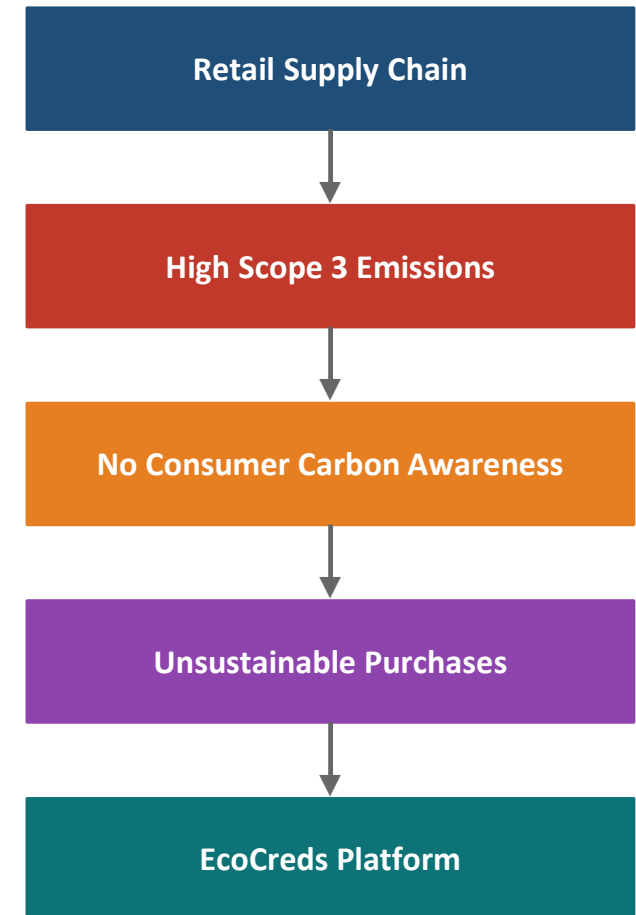
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Problem Statement / Our Theme

Retail with Purpose: Building a Sustainable and Responsible Future

- 1 Retail Scope 3 emissions account for >90% of a retailer's total carbon footprint (GHG Protocol)
- 2 Walmart likely to miss 2025–2030 climate targets — operational emissions rising (Reuters, 2024)
- 3 E-commerce platforms show price, ratings & delivery — but NOT environmental impact
- 4 Carbon labelling schemes are fragmented; many rely on broad category averages instead of product-specific data
- 5 No active consumer incentive to choose greener products — sustainability stays invisible at checkout



Problem Chain

Literature Review

Author(s)	Year	Key Contribution	Method	Limitation
Negi et al.	2014	Green SCM in Walmart; sustainable value networks & supplier programmes	Descriptive; secondary data	No consumer dimension; no verified carbon data
Mujtaba et al.	2007	Walmart global supply chain integration, management practices & tech adoption	Case study	Environmental aspects not explored
Vipin Jain	2021	RFID & robotics in Walmart/Amazon logistics for inventory optimisation	Technology review	Efficiency-focused; no carbon transparency
FasterCapital	2024	Environmental impact of large-scale retail; need for sustainability transparency	Sustainability analysis	No consumer-level carbon tracking solution
Reuters	2024	Walmart missing 2025 & 2030 climate targets; operational emissions rising	Investigative report	Corporate-level only; no consumer engagement focus

Research Gap: No existing study integrates product-level carbon intelligence into the consumer shopping workflow using AI-powered NLP + verified government emission datasets.

Dataset — DEFRA GHG Conversion Factors 2025

Source & Authority

The UK Government's DEFRA GHG Conversion Factors 2025 (Condensed Set) is the primary data source. Published annually by the Department for Energy Security & Net Zero, it provides peer-reviewed emission factors in kg CO₂e per kg of material — used by companies globally for official carbon reporting.

Why DEFRA?

- Government-verified — not AI-generated or unverified estimates
- Covers 16 material categories + 6 packaging categories
- Updated annually; widely used in official sustainability reports
- Used as the ground truth — Gemini NLP only maps materials TO DEFRA factors

Sample Emission Factors (kg CO₂e / kg)

Material	Factor (kg CO ₂ e/kg)
Virgin Plastic (PET/PP)	3.5 – 6.0
Recycled Plastic	1.5 – 2.5
Aluminium (virgin)	8.0 – 12.0
Aluminium (recycled)	0.5 – 1.0
Conventional Cotton	5.0 – 8.0
Organic Cotton	2.0 – 3.5
Polyester Fabric	5.5 – 7.0
Wood / Bamboo	0.1 – 0.5
Paper / Cardboard	0.7 – 1.2
Leather	14.0 – 20.0
Glass	0.8 – 1.2

Our Solution — EcoCreds

EcoCreds is a sustainability-first web platform that estimates product-level carbon footprints, recommends greener alternatives, and rewards eco-conscious purchasing — integrating DEFRA datasets with Google Gemini NLP.

DEFRA-Powered EcoScore

AI analyses product description; maps materials to DEFRA emission factors. Score 0–100 with Excellent / Good / Fair / Poor bands.

Smart Eco Swap Suggestions

Gemini picks the best lower-carbon alternative from the same category. Shows estimated CO₂ savings in kg.

EcoCreds Rewards System

3 pts / kg CO₂ saved + 1 pt / ₹100 spent + bonus pts for eco alternatives. Redeem for 5% Off coupon at 50 pts.

Offset Delivery Emissions

One-click carbon offset option at checkout making each order net-zero.

Price Tracker for Eco Products

Tracks and alerts for price drops on eco-friendly alternatives.

Secondhand Marketplace

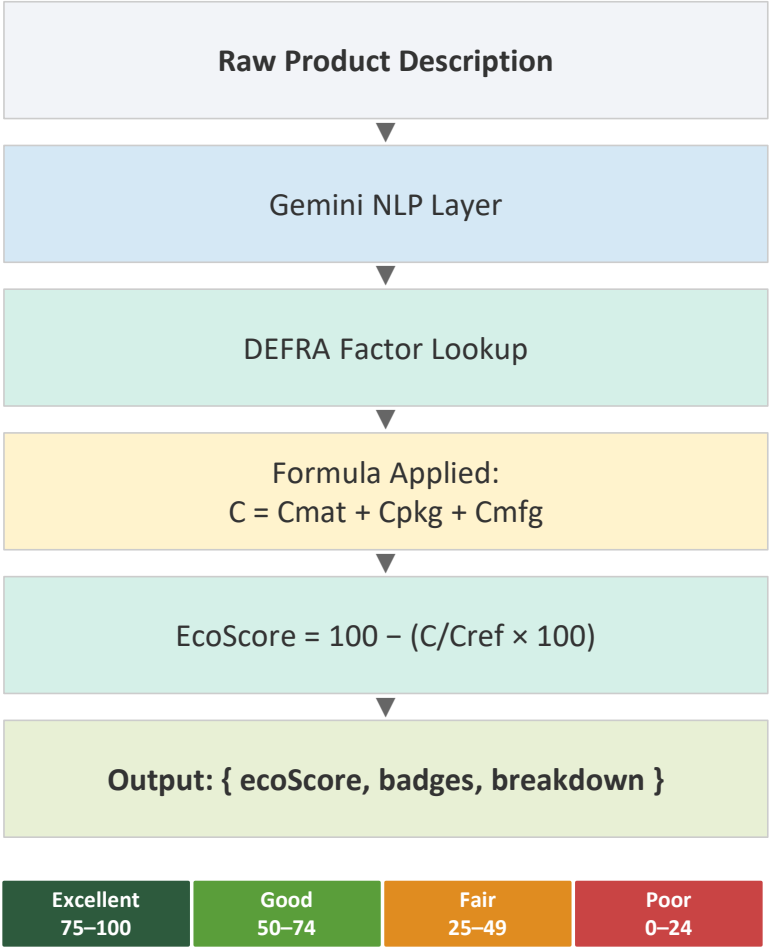
Peer-to-peer resale integrated in the same app. Seller earns +10 EcoCreds per sale — supports circular economy.

Methodology — Hybrid DEFRA + Gemini Carbon Engine

Existing Systems vs EcoCreds

Feature	Existing Systems	EcoCreds
Carbon Visibility	Not available to consumer	EcoScore per product (0–100)
Data Source	Category averages; inconsistent	DEFRA verified gov. factors
Consumer Incentives	None	EcoCred reward points + coupons
Greener Alternatives	Not suggested	AI-powered recommendation
Circular Economy	Separate platforms	Integrated secondhand marketplace
NLP Integration	Not used	Gemini NLP material extraction
Scope 3 Engagement	Supply-side only	Consumer-facing platform

Carbon Calculation Pipeline



Advantages / MVP

Drives Customer Engagement

EcoCreds introduces EcoPoints + Rewards that keep users returning and building sustainable habits long-term.

Achieves Scope 3 Green Goals

Promotes low-carbon product choices and lets customers offset emissions at checkout — directly reducing Scope 3 emissions.

Gamifies Green Behavior

Empowers customers as active sustainability participants through EcoScores, swap suggestions, and visible rewards.

Addresses Supply Chain Emissions

Encourages switching to locally sourced, low-impact products — reducing indirect emissions and transport footprint.

Improves Eco Brand Discoverability

Gives visibility to sustainable product lines; vendors in the sustainability programme benefit from increased ROI.

Builds Green Brand Image

Creates a positive sustainability experience without increasing product prices, logistics costs, or operational overhead.

Key Differentiators — Why EcoCreds Fits Any Retailer

Scope 3 Emissions Tackled at Consumer Level

Retailers struggle most with Scope 3 (>90% of carbon footprint). EcoCreds reduces these at the point of purchase — unlike back-office solutions, it is customer-facing and emissions-reducing at scale.

ESG Metrics Generated from Shopper Behaviour

Every EcoCreds checkout contributes measurable data (CO₂ saved, emissions offset, swaps made) pluggable directly into ESG reporting — built-in value from behaviour, not back-end estimation.

Empowers Green Suppliers as Discovery Engine

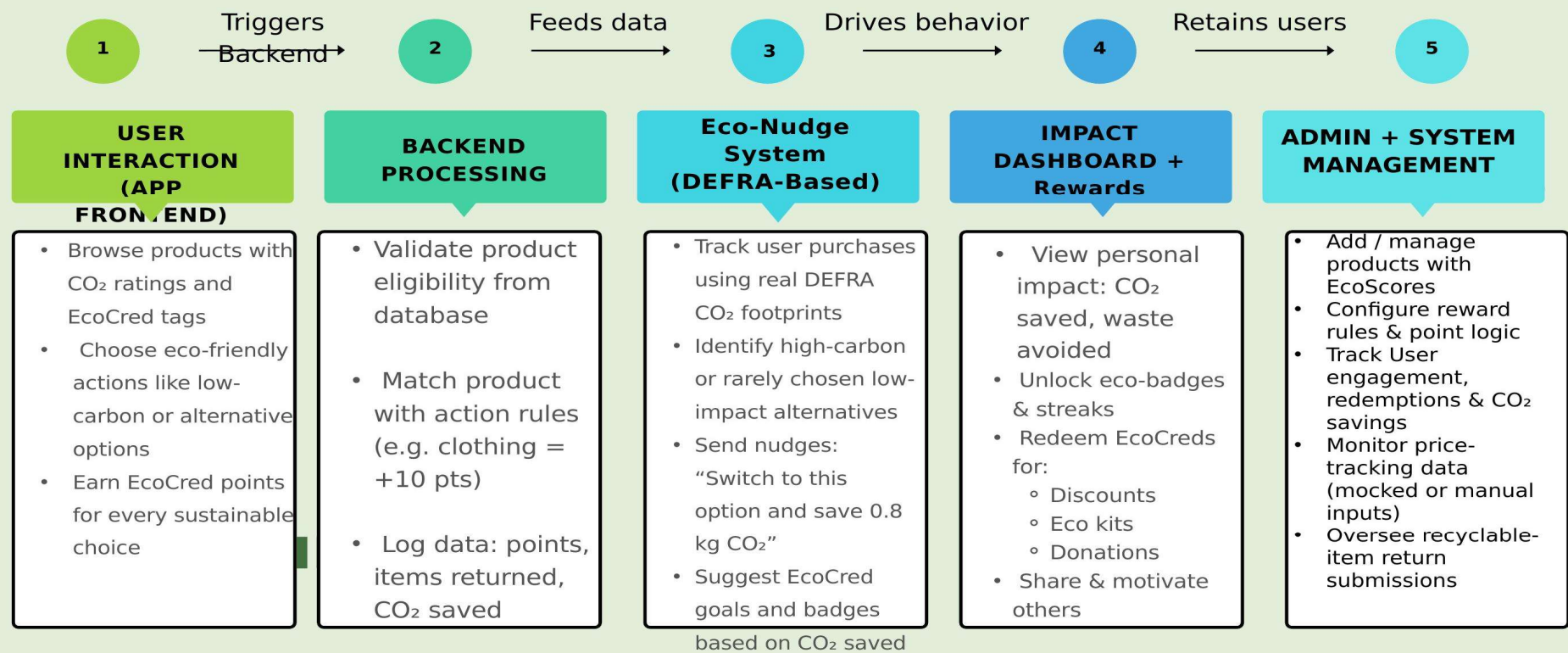
By surfacing eco-friendly alternatives, EcoCreds increases ROI for suppliers in the sustainability programme and creates market incentives for greener manufacturing.

Responsible AI — Verified Data, Not Hallucinated Values

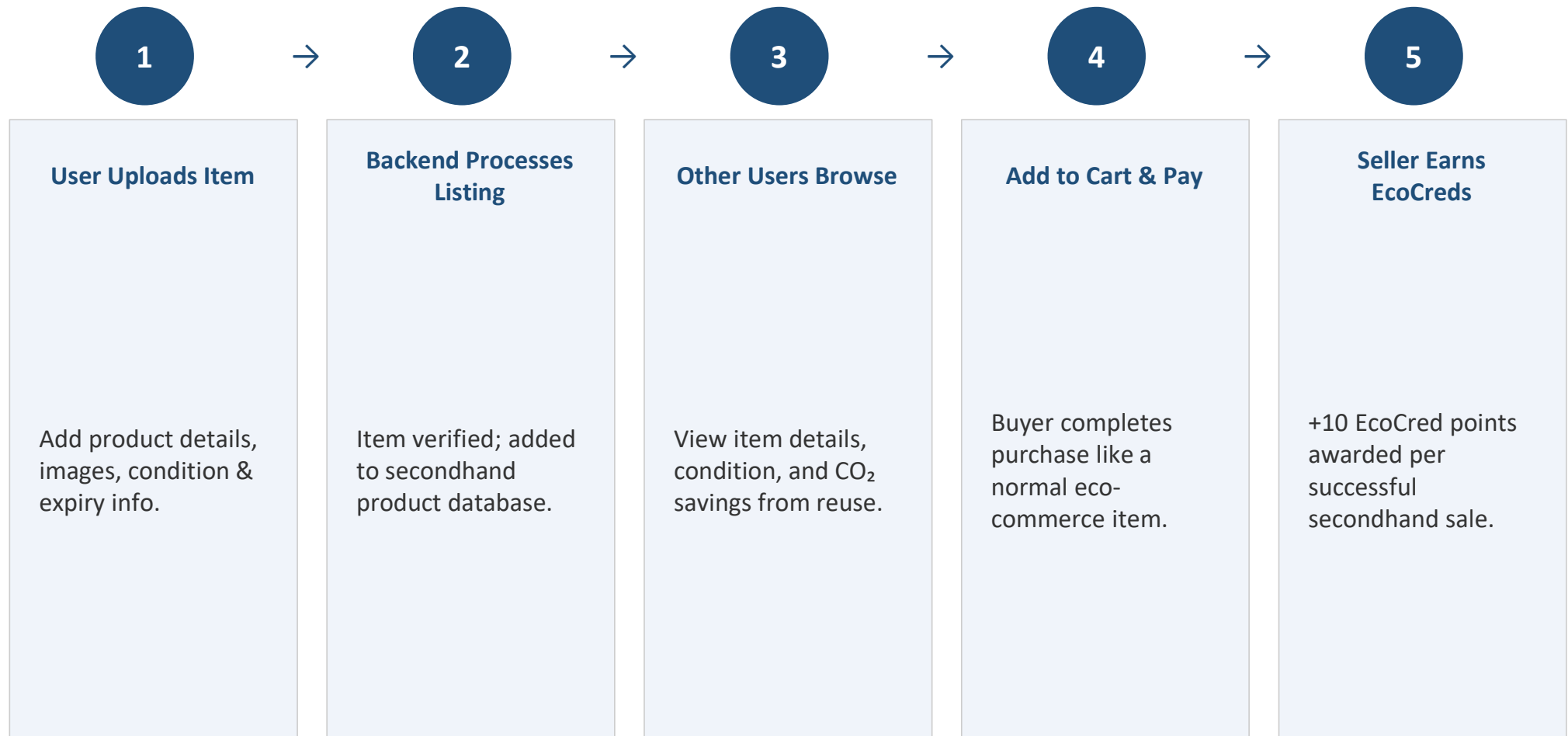
Gemini NLP is restricted to material extraction only. DEFRA 2025 dataset is the sole source of emission factors — ensuring scientifically grounded, auditable carbon calculations.

Process Flow 1 — Main App Flow

Process Flow 1

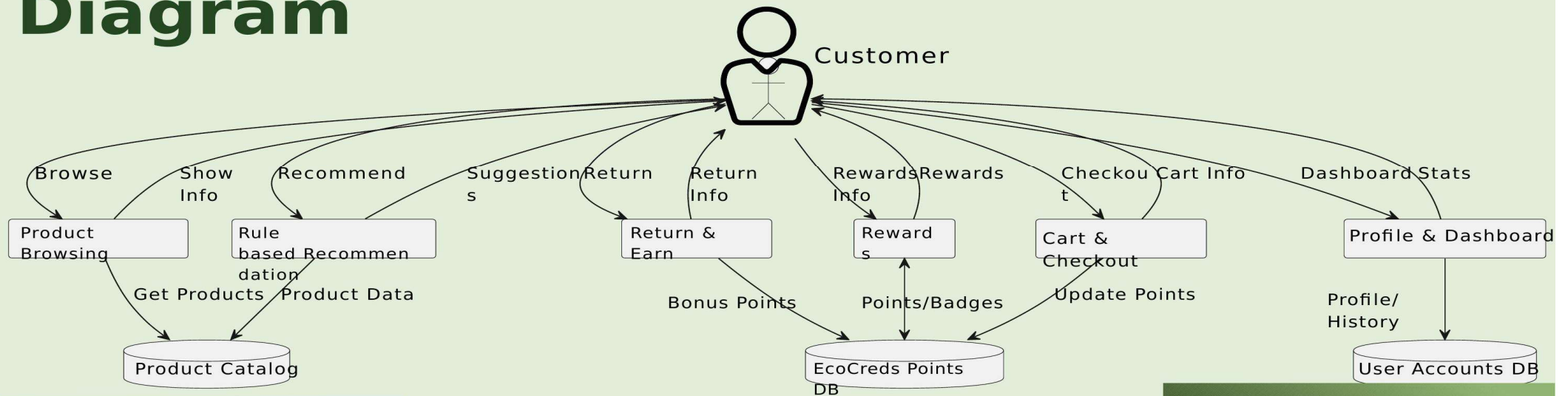


Process Flow 2 — Secondhand Marketplace



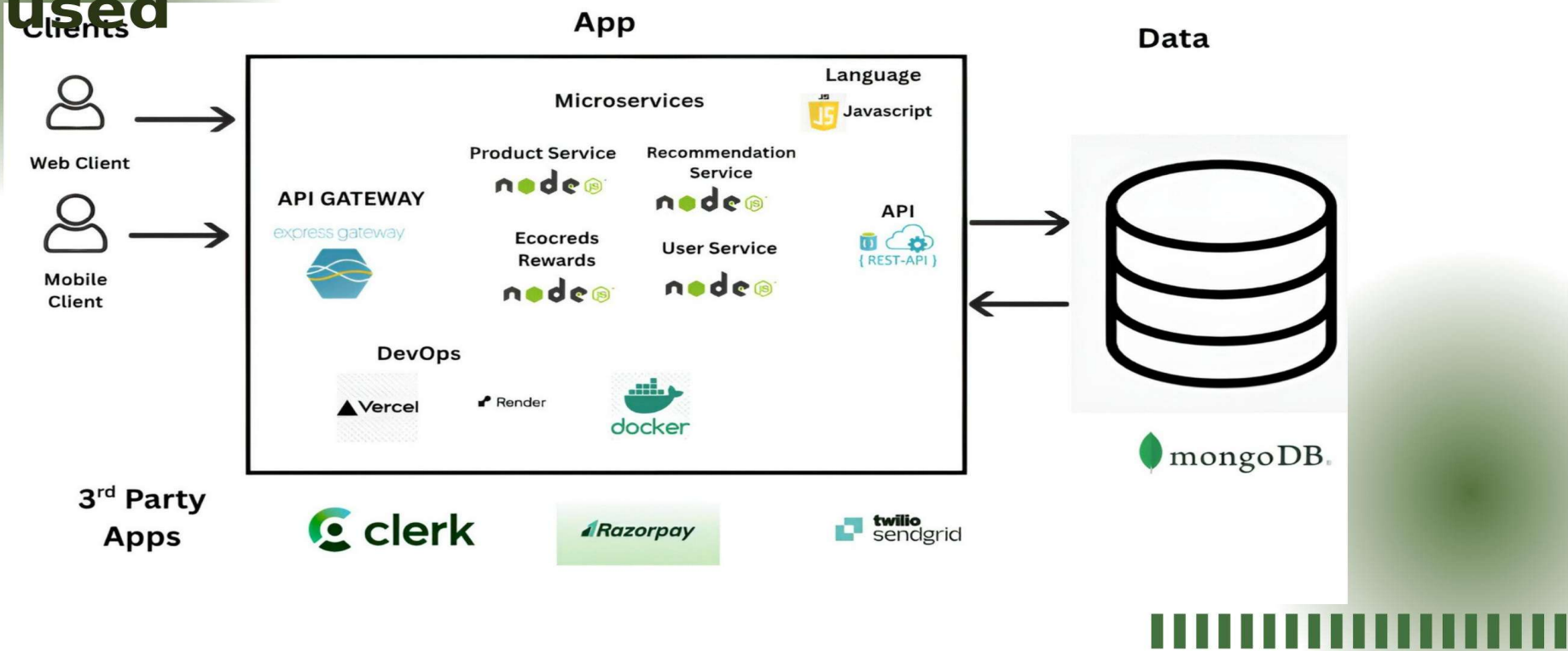
Use-Case Diagram

Use - Case Diagram



Technology Stack

Technology to be used



Results & Expected Outcomes

Expected EcoScore Distribution

Product Category	Band	Range
Recycled / biodegradable goods	Excellent	75–100
Natural fibre textiles	Good	50–74
Mixed-material electronics	Fair	25–49
Multi-layer packaged goods	Fair	25–49
Single-use plastic products	Poor	0–24

Recommendation Engine Outcomes

Scenario	Expected Outcome
Greener alternative found	Lower-carbon product shown with CO ₂ savings
No alternative available	Original retained; user informed
Avg. carbon reduction per swap	15% – 35%
Reward per kg CO ₂ saved	3 EcoCred points
Reward per secondhand sale	+10 EcoCred points

15–35%

Avg. CO₂ reduction
per eco swap

DEFRA 2025

Government-verified
emission source

Circular Economy

Secondhand
marketplace integrated

User Interface — Key Screens

Our Solution

EcoCreds is a sustainability-first shopping companion built for online shoppers.

It rewards customers for making eco-conscious choices and helps them track, reduce, and offset their carbon footprint — one cart at a time.

- **DEFRA dataset -powered EcoScore** on Every Product
- **Instantly rates a product's environmental impact** based on carbon footprint, material, packaging, etc.
- Smart Eco Swap Suggestions
- **Recommends greener alternatives based on less carbon footprint** — saves emissions before checkout.
- **EcoCreds Rewards System**
 - Customers earn points for making sustainable choices; points redeemable for coupons or donations (tree planting, etc.).
 - Offset Delivery Emissions at Checkout
- **One-click carbon offset option for delivery**, making orders net zero.
- **Price Tracker for Eco Products**
- **Tracks and alerts for price drops** on eco-friendly alternatives — saving money while saving the planet.
- Personal Sustainability Dashboard
- **Users see total CO₂ saved, rewards earned** and progress — drives continued engagement and

Product Browsing & EcoScore

Advantages/MVP

Drives Customer Engagement in Sustainability Journey

Empowers customers to be active participants in green mission — through visible EcoScores, swap suggestions, and rewards. This aligns with push for consumer-led sustainability.

Helps Achieve Green Project Goals

EcoCreds promotes low-carbon product choices and lets customers offset emissions at checkout — directly contributing to Scope 3 emissions reductions.

Gamifies Green Behavior

EcoCreds introduces EcoPoints + Rewards that keep users returning, choosing better, and building habits. This supports long-term goal of embedding sustainability into everyday behavior.

Addresses Supply Chain Emissions at the Consumer Level

EcoCreds encourages switching to locally sourced, low-impact products — reducing indirect emissions (transport, materials) and reinforcing supply chain transparency.

Improves Product Discoverability for Eco Brands & Suppliers

Gives visibility to sustainable product lines and encourages vendor participation.

Builds Green Brand Image

Creates a positive brand experience around sustainability, without increasing product prices, logistics costs, or operational overhead.

Advantages / Rewards Dashboard

Limitations — We're Thinking Long-Term

Incomplete or Inconsistent Product Data

Environmental metadata (materials, emissions, packaging) may not always be available for all products, reducing EcoScore accuracy. The system relies on structured product data — brands/sellers may need to improve reporting quality. A fallback baseline of 2.5 kg CO₂e/kg is applied when no DEFRA match is found.

Integration Complexity with Existing Systems

E-commerce platforms often use legacy or highly customised systems. Adding sustainability layers (EcoScore, alerts) requires careful staging to avoid breaking existing flows. A gradual rollout with A/B testing ensures smooth integration and minimal disruption.

Scope 3 Data Validation Challenges

Even with product-level estimates, supplier-side Scope 3 data is often incomplete or non-standard. EcoCreds may rely on DEFRA approximations until better lifecycle assessment data becomes widely available from suppliers.

Future Scope

Barcode / QR Scanning



Retrieve real-time EcoScores directly from physical retail products in-store.

Browser Extension



Overlay EcoScores on Amazon, Flipkart & Walmart product pages.

Mobile App



Dedicated Android & iOS app using React Native or Flutter.

Advanced Lifecycle AI



Supplier-specific data, transportation routes & manufacturing complexity for precise footprints.

Community Leaderboards



Social sustainability badges, peer-based engagement & monthly eco-reports.

Carbon Offset Integration



Let users neutralise unavoidable emissions using EcoCred points.

Blockchain Transparency



Verify product origin, material sourcing & sustainability claims on-chain.

Dedicated ML Model



Train a purpose-built sustainability classification model for carbon estimation.

Project endpoints and takeaways

Question	Answer
Project Accessibility for Next Batch	YES
Are you available as student mentor for this project to be taken further by next Batch	YES
Github repo link	https://github.com/ANT-TRIO/Ecocreds
Deployed link	https://eco-creds-prod.vercel.app/dashboard

References

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